

LongCat-DeepResearch

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ABSTRACT

We introduce LongCat-DeepResearch, which combines a LongCat model enhanced for deep research with a research harness organized around an executable ResearchSpec. Open-ended research requires investigation to adapt as analysis reveals new information needs. The harness shifts early iteration from full reports to a compact specification of research requirements, reducing reliance on repeated full-report rewriting. Multiple planners independently explore external sources, then consolidate and refine questions, evidence requirements, and section responsibilities. Researchers investigate and draft their assigned sections in separate contexts, gathering further evidence as their analyses develop; a Global Editor then coordinates local revisions to the assembled report. These stage interfaces also support research-task and trajectory construction for LongCat’s general-purpose model training. LongCat-DeepResearch achieves 55.25 on DeepResearchBench, 51.35 on DeepResearchBench II, and 79.83 on ResearchRubrics. It scores 76.04 on the in-house benchmark, ranking second among the four compared systems. Development-set analyses show that component and planning refinement effects vary across configurations and benchmarks, while enhanced Editor rounds improve average readability preference. These deep-research improvements are intended to be incorporated into the next general-purpose LongCat model release.



Figure 1: Public-benchmark scores.

1 Introduction

Open-ended deep research requires language-model agents to investigate underspecified questions and gather evidence from diverse external sources [Shao et al., 2024, Chen et al., 2024, Li et al., 2025]. These agents synthesize their findings into comprehensive long-form reports [Tongyi DeepResearch Team, 2025, Li et al., 2026c, Zhu et al., 2026b, Li et al., 2026b, Zhang et al., 2026b]. Unlike conventional information seeking, the evidence requirements of such reports cannot be fully specified before the investigation begins. As an analysis develops, emerging explanations expose missing evidence, unresolved claims require further verification, and new research questions arise. Research and writing must therefore inform one another throughout the investigation. This raises a fundamental question: *How can open-ended deep research adapt its investigation to information needs that emerge during writing?*

Existing systems instantiate this research–writing feedback in different ways. Prior approaches interleave reasoning, retrieval, and drafting [Li et al., 2025], or use an evolving report to guide subsequent retrieval and revision [Han et al.,

2025]. Other approaches organize investigation through multiple perspectives and evidence-grounded outlines before detailed composition [Shao et al., 2024, Li et al., 2026c]. When this interaction is carried by a growing research history or repeated full-report updates, three difficulties arise. First, evidence, intermediate reasoning, and draft text compete for finite context; compression may remove information whose relevance has not yet become apparent. Second, early findings can anchor the developing narrative and steer subsequent searches, narrowing the perspectives explored. Third, updating the full report repeatedly regenerates text beyond the passages affected by new evidence. The challenge is to retain feedback between research and writing without concentrating the entire investigation in an expanding draft.

In this work, we introduce LongCat-DeepResearch, a system combining an improved LongCat model with a research harness centered on an executable *ResearchSpec*. The key idea is to shift early iteration from the full report to a compact specification of its research requirements. Multiple planners independently search and read external sources to develop candidate specifications. Their proposals are consolidated and refined to identify missing questions, evidence requirements, and section responsibilities. This process encodes the emerging global understanding of the task into *ResearchSpec*, which serves as a compact proxy for what the report needs to establish. The downstream synthesis pipeline retains complete section drafts through assembly and editing. Each researcher receives the complete *ResearchSpec* and one assignment, continues gathering evidence in an independent context, and writes a complete section from its findings. The resulting section artifacts are assembled into a draft, after which a Global Editor identifies cross-section issues and Local Editors perform targeted revisions. This design combines compact global coordination with detailed local research and avoids repeatedly regenerating the full report as the investigation develops.

We evaluate LongCat-DeepResearch on three established benchmarks and an in-house benchmark. It achieves 55.25 on DeepResearchBench, 51.35 on DeepResearchBench II, and 79.83 on ResearchRubrics, outperforming the strongest of the three compared deep-research products by 0.30, 3.17, and 5.62 points, respectively, in the recorded system-level comparison (Section 4.3). On our in-house benchmark, LongCat-DeepResearch ranks second with an overall score of 76.04, exceeding Claude-DeepResearch at 61.42 and Gemini-DeepResearch at 42.49, and trailing ChatGPT-DeepResearch’s 76.59 by 0.55 points. Controlled analyses further show that additional planning cycles increase cost without consistently improving benchmark-native quality. The same stage interfaces also support the construction of research questions, task-specific rubrics, and trajectories used in the mid-training and post-training of LongCat’s general-purpose models.

2 Related Work

Context management and evidence preservation. Long-context access does not ensure reliable use of evidence throughout the input [Liu et al., 2023]. MemGPT separates working context from external storage [Packer et al., 2023], while AgentFold learns fine-grained context folding and RE-TRAC carries structured summaries across research attempts [Ye et al., 2025, Zhu et al., 2026c]. FoldAct studies the training instability introduced when generated summaries change future observations [Shao et al., 2025]. AdaCoM further shows that the preferred degree of compression varies with the underlying agent [Yi et al., 2026]. These works make compression an explicit, useful operation rather than an incidental implementation detail. In deep research, SearchSwarm returns compact, citation-grounded subagent reports to an orchestrator [Lan et al., 2026], and Argus synthesizes an answer from a compact evidence graph [Zhang et al., 2026c]. Deep-Reporter maintains a recurrent global summary and a verbatim local tail during sequential section generation [Ye et al., 2026]. Our report-synthesis path retains complete section artifacts through assembly and global-to-local editing. This interface carries complete section drafts into synthesis; external memory and compression address the separate management of research history.

Research planning and structured state. WebGPT and ReAct established tool-grounded interaction [Nakano et al., 2021, Yao et al., 2022]. STORM develops perspectives before article writing, and MindSearch decomposes search through an executable graph [Shao et al., 2024, Chen et al., 2024]. OmniThink uses an information tree and a conceptual pool to connect knowledge expansion with reflection [Xi et al., 2025]. More recent work makes the research structure mutable: WebWeaver interleaves evidence acquisition with outline optimization, AgentCPM-Report alternates drafting and deepening, and ScaffoldAgent selects outline changes using downstream utility [Li et al., 2026c,b, Yang et al., 2026]. Enterprise Deep Research combines coverage objectives, dependency-guided information sharing, and evidence-sufficiency conditions [Choubey et al., 2026]; SearchOS externalizes coverage, evidence, pending tasks, and failed searches [Zhang et al., 2026b]. DecomposeR makes a typed research DAG trainable with structure-aware rewards [Hussain et al., 2026].

The closest structured-state comparisons also include RhinoInsight, whose verifiable checklists constrain research actions and whose evidence audit links sources to drafted content [Lei et al., 2025], and DualGraph, which maintains separate, co-evolving knowledge and outline graphs [Shi et al., 2026]. These systems establish structured planning and evidence binding as existing design choices. In our pipeline, *ResearchSpec* defines section-level execution

responsibilities that are fixed after planning; researchers retain their findings in section artifacts passed to editorial reconciliation. Its role is an execution interface, rather than a persistent knowledge graph or an input-level specification for personalized query refinement [Yoon and Lee, 2026].

Long-form synthesis and revision. PaperQA2 develops cited scientific-topic summaries and literature-contradiction detection [Skarlinski et al., 2024]; OpenScholar combines literature retrieval, citation-backed generation, and iterative self-feedback [Asai et al., 2026]. These systems establish scientific synthesis as a research objective beyond isolated fact retrieval. In open-ended report writing, WebThinker interleaves reasoning, search, and drafting [Li et al., 2025], while TTD-DR uses an evolving draft to guide retrieval and iterative report revision [Han et al., 2025]. FS-Researcher separates persistent evidence collection from multi-session report writing and retains original source files [Zhu et al., 2026b]. CogGen coordinates a global plan–write–review loop with section-level work and permits global restructuring [Tian et al., 2026]; Ptah maintains inspectable research artifacts and visual memory for multimodal composition [Zhang et al., 2026a]. Our design assembles the independently written sections before deciding how to reconcile them: the Global Editor specifies ownership, while Local Editors revise assigned units. This scope differs from regenerating the entire document in one invocation, but does not guarantee that edits retain every useful detail. Mr. Dre provides direct evidence that report revision can satisfy new feedback while damaging earlier coverage or citation quality [Chen et al., 2026]. DeepTRACE audits statement-level support and attribution, distinguishing listed sources from supported claims [Venkit et al., 2025]. Self-Refine, DuMate, and AREX offer complementary feedback and verification loops [Madaan et al., 2023, Yan et al., 2026, Lu et al., 2026]. Dependency-aware replay is a possible extension of the section interface.

Synthetic research tasks and trajectories. OpenAI’s Deep Research System Card describes reinforcement learning on browsing tasks that include open-ended tasks graded with rubrics [OpenAI, 2025]. DR Tulu develops an open long-form research training approach in which rubrics evolve with the policy and newly acquired evidence [Shao et al., 2026]. These are distinct precedents: rubric-based supervision predates the evolving-rubric formulation. Tongyi DeepResearch and Step-DeepResearch describe synthetic tasks and agentic training across mid-training and post-training [Tongyi DeepResearch Team, 2025, Hu et al., 2025]. S1-DeepResearch emphasizes planning, evidence integration, and report generation beyond search-centric QA [Dong et al., 2026], while Marco DeepResearch emphasizes verification in task and trajectory construction [Zhu et al., 2026a]. For open-ended reports, requirements must be supported and aligned with the question. ResearchRubrics contributes expert-written prompts and criteria, including implicit requirements and negative criteria [Sharma et al., 2025]; DeepResearchBench II derives atomic criteria from expert articles [Li et al., 2026a]. These evaluation resources are distinct from synthetic training data. DeepRubric constructs evidence trees and jointly synthesizes queries and rubrics, whereas Quest uses rubric trees to construct both objective and open-ended research tasks [Zhu et al., 2026d, Xie et al., 2026]. Our data discussion builds on these principles and describes document provenance, question–rubric alignment, and trajectory interfaces. It does not introduce a new optimization algorithm or attribute the evaluated model’s performance to one training stage.

3 Method

The problem is to sustain evidence gathering and report construction when each model call has finite effective context. LongCat-DeepResearch addresses this by assigning section research to independent contexts and separating global editorial decisions from local text generation. Its intermediate artifacts also provide units for data construction and stage-wise evaluation.

3.1 Research Harness

ResearchSpec assigns research responsibilities before detailed evidence accumulates. Independent Researchers develop the assigned sections; assembly and editing then coordinate those sections into a report (Fig. 2). Retaining these section artifacts does not imply preserving every retrieved page or every detail of the underlying interactions.

Explore before committing to a plan. Independent Planning Writers (planners) briefly search and read sources before proposing the report’s structure, grounding the research scope in available evidence beyond the model’s prior knowledge. Their separate explorations can reveal complementary perspectives, as in perspective-guided article planning [Shao et al., 2024]. The Planning Judge consolidates the candidate specifications, the Critic searches for missing questions and cases, and the Reviser incorporates the feedback. Because section assignments direct subsequent research, this refinement seeks to identify omissions before a direction is left uninvestigated. Additional candidates and refinement rounds provide ways to allocate more computation to planning; their benefit is not guaranteed.

ResearchSpec records each section’s scope, research questions, required entities or cases, and provisional source leads. This compact representation of the intended report also specifies how research work is divided across contexts. Its

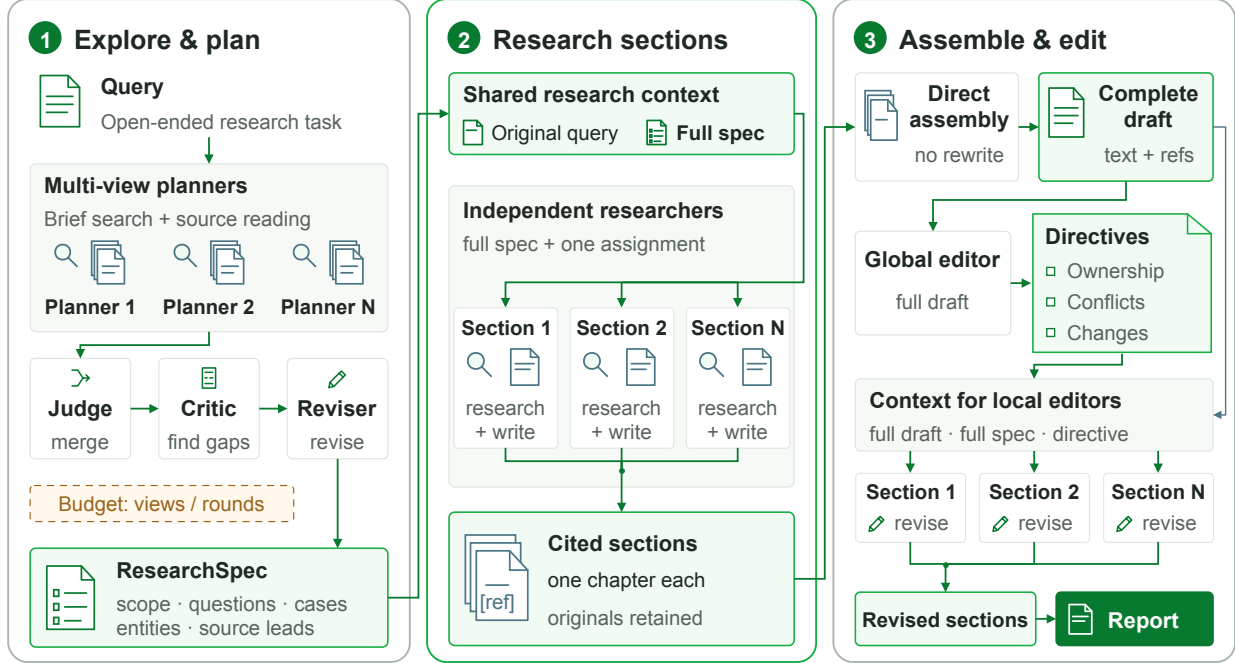


Figure 2: The LongCat-DeepResearch harness.

coverage and responsibilities can be inspected and revised before generating a full draft. Each dispatchable subsection has a unique hierarchical ID, such as S1.2; we refer to these execution units as sections. Before dispatch, validators check ID uniqueness, parent relationships, consecutive ordering, and required fields. Invalid plans undergo repair or revert to a validated planning candidate; unresolved validation errors stop dispatch. Findings and source leads must still be established through subsequent research and verification. The specification is revised within the planning stage and then held fixed during section research; a later research round that reopens the plan is an extension beyond the current pipeline.

Research and write in independent section contexts. Each Researcher receives the original query, the complete ResearchSpec, and one section assignment. The global specification establishes how its local work contributes to the report, while separate contexts accommodate the branches’ detailed tool interactions. Researchers receive the shared specification, without depending on previously generated sections, and can extend their investigation while addressing the assigned requirements. They execute concurrently and write their own citation-bearing sections from the evidence they have acquired. The same agent therefore carries its local evidence into writing without first compressing it into a summary for a separate writer. The complete sections retain the developed arguments and details, reducing the reliance of subsequent composition on a repeatedly compressed shared history.

Assemble first, then coordinate and edit. The harness mechanically assembles sections in ResearchSpec order, retaining their text and citations even when content overlaps. A Global Editor reads the complete draft, assigns ownership of repeated material, identifies conflicts, and specifies changes. Local Editors apply these directives to their assigned sections using the complete draft and specification as context, drawing on facts and citations already present in the draft. Their output scope is local, so one editorial call need not regenerate the entire document. This separates the global judgment needed for coordination from the generation of revised text, while keeping the original sections and assembled draft available for comparison. Both editor types still read the full draft and remain subject to input-context limits; research and editing can still omit useful details.

Decoupled interfaces for data construction and evaluation. Planning maps the query and explored sources to a ResearchSpec; research maps section assignments to grounded sections; editing maps a draft and directives to revised sections. These mappings provide concrete units for targeted data synthesis and rubric-based evaluation. ResearchSpecs can be checked for coverage before detailed investigation, sections against their assigned requirements, and revisions against editorial directives and original text. The decomposition supports separate improvement and computation allocation at each stage, and provides the foundation for the research-data construction described next.

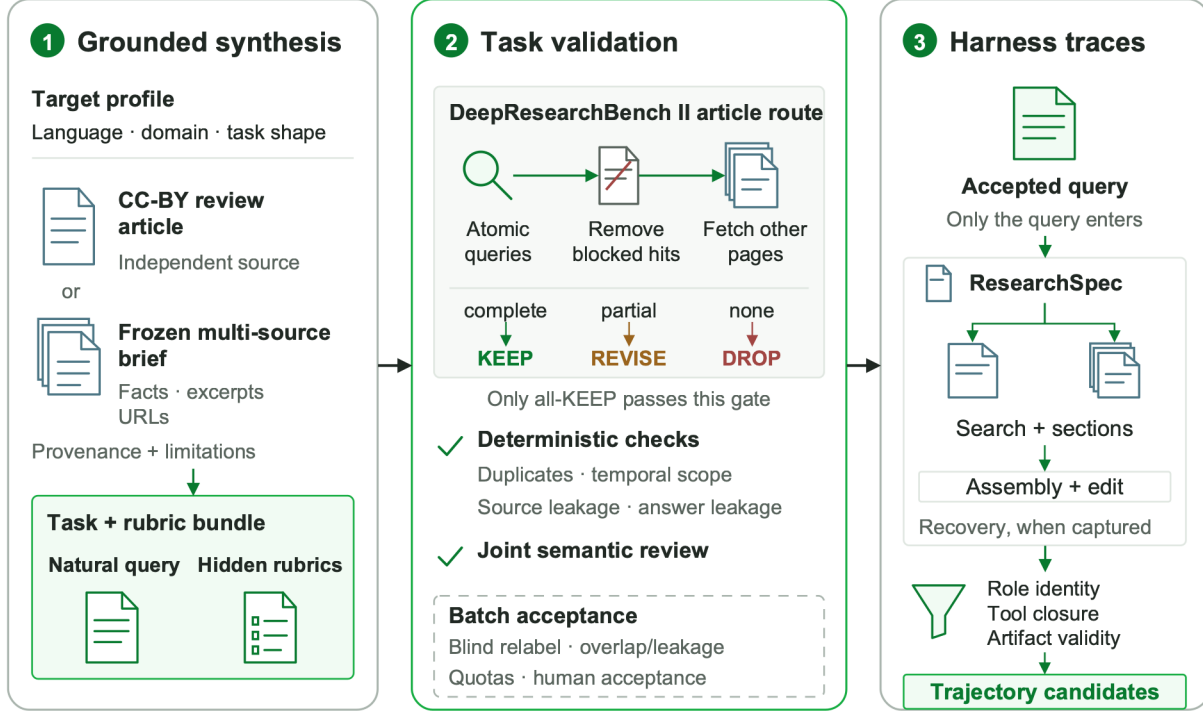


Figure 3: Evidence-grounded construction of research tasks and trajectories.

3.2 Research Data Construction

Data construction follows the harness’s stage interfaces: we first build research tasks with evidence-backed requirements, then collect trajectories that show how those tasks are planned, investigated, and edited. Independent source material grounds the questions and rubrics used in this process.

Ground questions and rubrics in source material. A target profile specifies language, topic, breadth, and the intended report. The article-based route starts from an independently licensed review article; a complementary route uses frozen multi-source briefs containing facts, excerpts, URLs, and source limitations. Both routes organize evidence for questions that require explanation or comparison across sources.

Each question is paired with task-specific rubrics. Following evidence-grounded synthesis principles [Zhu et al., 2026d, Xie et al., 2026], factual criteria must have supporting evidence, while analytical criteria specify warranted comparisons or inferences. The rubric can include implicit requirements and negative conditions, as distinguished in ResearchRubrics [Sharma et al., 2025]. Construction evidence and rubrics remain separate from the query supplied to the answering agent.

Check evidence, searchability, and task quality. The article-based route supports a bounded searchability check for atomic information-recall criteria. It searches for alternative sources, removes hits from the excluded construction article, fetches the remaining pages, and checks support for the complete factual requirement. Complete, partial, or missing support produces **KEEP**, **REVISE**, or **DROP**; the gate passes only when all retained criteria receive **KEEP**. Deterministic checks address duplicate rubrics, answer leakage, excluded-source leakage, and temporal scope. Joint semantic review checks whether the question, evidence, and rubric describe a coherent research task. Batch records identify which checks were executed and whether their acceptance conditions were met. The bounded search tests evidence availability within its budget, without certifying exhaustive Web answerability.

Collect and filter stage-specific trajectories. Accepted queries drive teacher executions of the harness, recording candidate and final ResearchSpecs, tool requests and observations, citation-bearing sections, assembled drafts, and editorial decisions. Parallel branches retain their actual inputs and dependencies. Filtering checks role identity, tool-call/response closure, and intermediate-artifact validity, while preserving distinctions among complete, partial, and failed attempts. Versioned provenance binds each record to its teacher and harness, so that outputs remain attributable

to the execution that produced them. Dataset-level overlap checks, distribution-aware selection, and human acceptance where specified remain separate release gates. A candidate trajectory does not itself establish use in a trained checkpoint.

Model training. Research-related data are used alongside other data in the mid-training and post-training of LongCat’s general-purpose models to improve deep-research capabilities. LongCat-DeepResearch combines the improved LongCat model with the research harness described in [Section 3.1](#); it is not a separately released specialist model. These improvements are intended to be incorporated into the next general-purpose LongCat model release. We retain the evaluated checkpoints’ identities in the experimental record; the broader training mixture and optimization recipe belong to the model technical report. System-level scores do not isolate the contribution of one data source or training stage.

4 Evaluation

We organize evaluation around four benchmarks: DeepResearchBench, DeepResearchBench II, ResearchRubrics, and an in-house benchmark. Following each benchmark’s official evaluation setup, DeepResearchBench RACE and DeepResearchBench II are judged by GPT-5.5 (medium), while ResearchRubrics is judged by Gemini 2.5 Pro. These models serve as evaluators of the generated reports. Evaluator-specific reference results are available from the [official benchmark leaderboard](#). The official snapshots and the four systems evaluated here are presented together in [Section A](#). The first three provide the recorded comparison with Gemini-DeepResearch, ChatGPT-DeepResearch, and Claude-DeepResearch. The in-house benchmark is reported in [Section 4.5](#). The intended protocol supplies the original task prompt without directly providing benchmark rubrics or reference reports. This does not prevent retrieval from the public Web.

4.1 Benchmarks

DeepResearchBench. DeepResearchBench contains 100 PhD-level research tasks spanning 22 domains, balanced between 50 Chinese and 50 English tasks [\[Du et al., 2025\]](#). Its RACE protocol assesses each report along four fixed top-level dimensions: *Comprehensiveness*, *Insight/Depth*, *Instruction-Following*, and *Readability*. RACE generates task-specific criteria and weights within these dimensions, scores both a target and a strong reference, and computes the final relative score as $100 \times S_{\text{tgt}} / (S_{\text{tgt}} + S_{\text{ref}})$ on the percentage scale, where S_{tgt} and S_{ref} denote the corresponding target and reference scores. Relative scores are computed within each task before macro-averaging across tasks. We report dimension scores where they are available; the current LongCat-DeepResearch row includes all 100 tasks from the September 14 evaluation. For this run, we use the benchmark’s GPT-5.5 (medium) evaluator branch with its official prompt, dynamic criteria, reference reports, weights, and aggregation code unchanged. The reference reports are the April 2025 snapshot generated by Gemini-DeepResearch. DeepResearchBench also provides FACT, which extracts statement–URL pairs, removes duplicate claims associated with the same URL, and checks whether the retrieved page supports each statement. The official current pipeline uses GPT-5.4 Mini for extraction and support judgment and Jina Reader to obtain page text. These quantities should be accompanied by output length because effective-citation counts are length-sensitive. We do not report FACT results. FACT measures statement–page support; it does not by itself establish source authority or the truth of every claim.

DeepResearchBench II. DeepResearchBench II contains 132 tasks across the same 22-domain taxonomy, with 66 tasks in each language [\[Li et al., 2026a\]](#). The benchmark is grounded in expert-written investigative articles. The benchmark paper reports 9,430 atomic rubrics constructed through automatic extraction, self-evaluation, manual revision, and more than 400 hours of expert review. Its rubrics cover *Information Recall*, *Analysis*, and *Presentation*. A rubric passes only when the report contains the required fact or inference; numerical requirements must be matched explicitly. We use the official evaluator and report the pass rate for each dimension together with the official overall score (the mean of task-level pass fractions across all rubrics). Task rubrics and reference text are not deliberately supplied in the actor prompt. If a supporting sentence cites a blocked source article, the corresponding rubric receives the official score of -1 ; we report the blocked rate in addition to pass rates. The product comparison uses GPT-5.5 (medium) as the judge.

ResearchRubrics. ResearchRubrics pairs 101 realistic prompts with 2,593 expert-written criteria covering explicit and implicit requirements, synthesis, citation quality, instruction following, and communication quality [\[Sharma et al., 2025\]](#). We compute a weighted score per task before macro-averaging across tasks. The LongCat result covers all 101 tasks. Dimension scores use the sum of score times weight in the numerator and only positive weights in the denominator; negative-weight criteria contribute penalties to the numerator. Dimension scores are not averaged to obtain the overall result. We use Gemini 2.5 Pro as the judge, following the benchmark’s evaluation setup, and retain its

original expert-written criteria and weighted scoring convention. This benchmark complements DeepResearchBench II by testing adherence to prompt-specific criteria that include both requested content and requirements inferred from task context.

In-house benchmark. The in-house comparison evaluates LongCat-DeepResearch, Gemini-DeepResearch, ChatGPT-DeepResearch, and Claude-DeepResearch on the in-house benchmark. GPT-5.6 Sol (xhigh) scores the reports with query-specific pointwise rubrics and bidirectional pairwise comparisons. The resulting dimension scores and weighted overall results are reported in [Section 4.5](#).

4.2 Comparison Systems

The comparison includes LongCat-DeepResearch, Gemini-DeepResearch, ChatGPT-DeepResearch, and Claude-DeepResearch. The latter three are evaluated through their native research products, including their tools and budgets. Their model snapshots are Gemini 3.7 Flash, GPT-5.6 Sol (xhigh), and Claude Opus 5 (xhigh), respectively. We use the product names in result tables and the full model names when describing experimental configurations. These system-level comparisons do not isolate a model or harness component. The configuration study in [Section 5.4](#) separately compares the previous LongCat release with ReAct/direct writing and the current harness, and compares that previous release with the current LongCat model under the same current harness.

4.3 Evaluation Protocol

The benchmark task sets contain 100 DeepResearchBench tasks, 132 DeepResearchBench II tasks, and 101 ResearchRubrics tasks. Scored coverage and missing-task policies are recorded in [Section B](#). DeepResearchBench and DeepResearchBench II use GPT-5.5 (medium), and ResearchRubrics uses Gemini 2.5 Pro. LongCat-DeepResearch uses a 300k-character DeepResearchBench II judge-input limit, while the other compared systems use 150k. Per-benchmark execution details are reported in [Section B](#). The results are point estimates; no public leaderboard rank or statistical-significance claim is made.

4.4 Main Results

[Table 1](#) reports the current LongCat-DeepResearch comparison. LongCat-DeepResearch obtains 55.25 on DeepResearchBench, 51.35 on DeepResearchBench II, and 79.83 on ResearchRubrics. Relative to the strongest of the three listed web products, these are differences of +0.30, +3.17, and +5.62 points, respectively. LongCat-DeepResearch has the highest overall scores on these three public benchmarks among the compared systems.

DeepResearchBench dimensions. LongCat-DeepResearch scores 55.25, 0.30 points above ChatGPT-DeepResearch’s 54.95. Comprehensiveness, insight, instruction following, and readability are 56.21, 56.48, 55.18, and 49.96, respectively. All five LongCat scores are macro-averaged over the same 100 tasks.

DeepResearchBench II. The selected run scores 51.35, with information recall 46.42, analysis 60.69, presentation 83.13, and a blocked rate of 0.79%. Its 3.17-point margin over Claude-DeepResearch is reported with the 300k/150k judge-input-limit difference specified in the table and [Section B](#).

ResearchRubrics. LongCat-DeepResearch scores 79.83 over all 101 cases, 5.62 points above ChatGPT-DeepResearch’s 74.21. The dimension scores in [Table 1](#) use the positive-weight denominator. LongCat has the strongest displayed scores for explicit requirements, implicit requirements, and synthesis; instruction following and communication remain weaker than at least one listed web product. The macro-average result cannot be reconstructed by simply averaging these dimension scores. The web-product rows use their recorded scoring cohorts ([Section B](#)).

[Section C](#) follows a high-scoring and a low-scoring DeepResearchBench II report through their actual ResearchSpec, section drafts, and editorial changes.

4.5 In-house Benchmark

We compare three leading international deep-research systems and LongCat-DeepResearch on the in-house benchmark. GPT-5.6 Sol (xhigh) evaluates each report using query-specific pointwise rubrics and bidirectional pairwise comparisons. The In-house block of [Table 1](#) reports Task, Evidence, Analysis, Synthesis, and Organization/Expression scores together with the weighted overall score. The report-order protocol is summarized in [Section B.1](#).

Metric	Gemini-DeepResearch	ChatGPT-DeepResearch	Claude-DeepResearch	LongCat-DeepResearch
DeepResearchBench				
Comprehensiveness	49.47	55.41	53.71	56.21
Insight / Depth	50.73	55.82	53.87	56.48
Instruction following	50.88	54.99	53.59	55.18
Readability	49.28	51.51	51.35	49.96
Overall	50.21	54.95	53.43	55.25
DeepResearchBench II				
Information recall	41.75	41.97	44.37	46.42
Analysis	51.28	52.26	51.15	60.69
Presentation	84.81	86.98	81.02	83.13
Blocked rate	1.62%	0.00%	0.89%	0.79%
Overall	46.72	47.16	48.18	51.35
ResearchRubrics				
Explicit requirements	76.36	83.57	79.64	86.97
Implicit requirements	60.25	70.36	70.57	79.52
Instruction following	62.39	74.47	74.22	66.27
Citation quality	32.71	64.43	67.27	65.00
Synthesis	60.41	69.14	70.29	78.92
Communication	59.64	64.24	53.33	60.00
Overall	64.92	74.21	72.91	79.83
In-house				
Task	81.64	92.14	88.25	93.95
Evidence	41.72	71.98	58.59	66.56
Analysis	20.66	69.59	42.34	67.42
Synthesis	16.57	63.57	45.80	74.06
Organization / Expression	21.35	74.64	53.20	52.99
Overall	42.49	76.59	61.42	76.04

Table 1: Public and in-house benchmark results with benchmark-specific dimensions. In-house Overall is the weighted total; bold marks the highest in-house score for each metric. Higher is better except for blocked rate. Scores use each benchmark’s percentage-scale convention. LongCat uses a 300k-character DRB-II evaluator-input limit; the compared web products use 150k. Scored coverage differs across systems (Section B).

5 Design Analysis

We study component ablations, ResearchSpec refinement, Editor scaling, and model-harness configurations on fixed development subsets. Model and harness changes are explicit interventions within each study; cohorts and protocols are reported separately, and scores are not pooled across studies. DRB-I, DRB-II, and RR abbreviate DeepResearchBench, DeepResearchBench II, and ResearchRubrics, respectively.

5.1 Component Ablations

Table 2 compares the complete pipeline (Full) with three component interventions for LongCat. Full uses three Planning Writers, the Planning Judge/Critic/Reviser, parallel Researchers, and the Editor. One Writer → Reviser runs these two stages in sequence and removes the Planning Judge and Critic; it therefore changes the planning procedure as well as Writer count. One whole-report Researcher receives Full’s complete ResearchSpec but uses a single research history to produce the draft. No Editor uses the exact draft saved before editing in its originating Full run.

All conditions within each displayed column use identical question IDs. The LongCat DeepResearchBench II comparison uses the common completed questions across all four arms. Selection and the stopped case are documented in Section B.2.

Configuration	DRB-II	ResearchRubrics
Full	48.68	79.14
One Writer → Reviser	44.65	74.46
One whole-report Researcher	45.64	78.49
No Editor	48.60	78.00

Table 2: LongCat component scores; higher is better. DRB-II denotes DeepResearchBench II, and → denotes stage order. Differences in the text are computed before rounding. Bold marks the highest mean per column. Detailed configurations are given in [Section B.2](#).

For LongCat, Full has the highest observed mean in both displayed subsets. Simplifying Planning lowers the means by 4.04 points on DeepResearchBench II and 4.68 on ResearchRubrics. A single Researcher lowers them by 3.04 and 0.65 points, respectively. The Editor differences are smaller: Full exceeds its unedited draft by 0.08 and 1.14 points.

These are previously inspected development subsets with unequal realized computation and different retrieval windows. Run provenance and recovery details are documented in [Section B.2](#). The table reports native scores; dedicated fact-support and readability evaluations are not complete for these four-condition cohorts.

5.2 ResearchSpec Refinement

We examine how planning aggregation and refinement affect final reports on a fixed ResearchRubrics development subset with LongCat. W0 uses a fixed single Writer’s ResearchSpec; J0 integrates three Writers through the Judge, before any Critic/Reviser cycle. R1 and R2 apply one and two successive Critic/Reviser cycles to J0. Each stage runs the same downstream Researcher and Editor configuration; planning compaction is disabled.

[Table 3](#) pairs a fast ResearchSpec coverage diagnostic with final-report ResearchRubrics scores for explicit requirements, implicit requirements, and the full-rubric total. The coverage diagnostic evaluates the plan without running Researcher or Editor. Moving from W0 to J0 improves coverage and all three report scores. R1 further improves explicit-requirement scores, while implicit-requirement scores decrease. The highest total occurs at J0. LongCat’s coverage peaks at R1, illustrating that more complete planned requirements need not yield the highest-scoring final report. The protocol and subscore definitions are given in [Section B.4](#).

Stage	ResearchSpec	ResearchRubrics		
	Coverage (%)	Explicit	Implicit	Total
W0	53.76	84.34	82.21	81.67
J0	60.47	89.09	87.11	85.13
R1	63.14	91.59	82.76	84.34
R2	62.48	90.91	87.99	84.89

Table 3: LongCat planning and report scores on a fixed ResearchRubrics development subset. W0 is the single-Writer plan, J0 the Judge-integrated plan, and R1/R2 one/two Critic/Reviser cycles. ResearchSpec coverage evaluates the plan before report generation. RR Explicit and Implicit score the corresponding report-rubric categories; Total uses the full rubric. Bold marks the best stage for each metric. The protocol is given in [Section B.4](#).

5.3 Editor Scaling

We study LongCat Editor scaling on fixed development subsets for each benchmark, with comparisons matched by question. A0 is the unedited draft; A1 applies the original Editor once. B1 starts from A0 with an enhanced Editor using source excerpts and fact-checking instructions. B2 and B3 continue the same procedure for two and three rounds, respectively ([Section B.6](#)). [Table 4](#) reports native scores and average readability preference for these conditions.

Average readability preference. Across the enhanced Editor rounds B1–B3, the mean preference rises from 50.42 to 53.33 to 53.96 for LongCat. A1 is a separate original-Editor control, not an earlier enhanced round. These averages describe organization and expression preferences across the two benchmarks; individual benchmark trends differ.

Native benchmark scores. The three-round enhanced Editor (B3) raises DRB-II from 55.87 to 58.38 relative to A0; ResearchRubrics is nearly unchanged, from 85.91 to 86.00. The enhanced intervention jointly changes source access,

Configuration	DRB-II	RR	Readability preference (avg.)
A0: No Editor	55.87	85.91	—
A1: Original Editor (1 round)	56.04	85.66	50.00
B1: Enhanced (1 round)	55.15	85.18	50.42
B2: Enhanced (2 rounds)	55.68	86.87	53.33
B3: Enhanced (3 rounds)	58.38	86.00	53.96

Table 4: LongCat Editor scaling. DRB-II (DeepResearchBench II) and RR (ResearchRubrics) are native report scores on the development subsets. Readability is the mean automatic preference score across the development subsets against the same A0 draft; 50 denotes parity. A dash marks A0 as the ungraded reference. Bold marks the highest mean per column, without implying statistical significance.

instructions, and editing computation, so it does not isolate a factual-verification effect. Dedicated evidence-support evaluation remains unavailable. Full paired results and protocol details appear in [Section B.6](#).

5.4 Model and Harness Configurations

We compare three configurations using benchmark-specific evaluation cohorts. The first uses the previous LongCat release with bounded ReAct research followed by a direct report. The second pairs the same previous release with the current harness. The third pairs the current LongCat model with the current harness. ReAct uses the same search and page-reading services and a 65,536-token final-output ceiling, with thinking enabled.

[Table 5](#) reports native report quality for the three configurations.

Configuration	DRB-I	DRB-II	RR
LongCat (previous release) + ReAct / direct report	47.65	33.13	60.33
LongCat (previous release) + current harness	53.58	48.68	71.90
LongCat-DeepResearch	55.25	51.35	79.83

Table 5: Report quality across model and harness configurations. Previous release refers to LongCat-2.0. The LongCat-DeepResearch row reports the full-cohort main results from [Table 1](#); the other rows use matched evaluation subsets.

The displayed differences between LongCat-DeepResearch and the previous release with ReAct/direct writing are 7.60 points on DeepResearchBench, 18.22 on DeepResearchBench II, and 19.50 on ResearchRubrics. Relative to the previous model release under the current harness, the differences are 1.67, 2.67, and 7.93 points, respectively. Because the first two rows use matched subsets while the LongCat-DeepResearch row uses the full cohorts, these are descriptive configuration comparisons rather than strict paired estimates. Generation uses bounded retries and format recovery with original attempts and costs retained; these results do not estimate first-attempt reliability or matched-compute efficiency.

5.5 Case Study: From ResearchSpec to Report

We examine a recorded LongCat trajectory for *AI-enhanced portfolio management* (DeepResearchBench II 012). The user requests a review of classical theories, AI/ML applications, multi-criteria decision making, and portfolio optimization and rebalancing, covering methods through April 2024. The case exposes two concrete operations: revising the research agenda before section research, and coordinating overlapping evidence after the section drafts have been assembled. We follow the recorded artifacts and quote their original wording rather than reconstructing an idealized run.

5.5.1 Turning the Question into Research Assignments

The merged ResearchSpec decomposes the four requested themes into 19 subsections: ten for classical theories, five for AI/ML applications, two for multi-criteria methods, and two for optimization and rebalancing. Each subsection contains *What to Cover*, research questions, required entities or cases, and source leads. These fields separate the topic of a section from the evidence that its Researcher must seek. The specification also records shared boundaries, including the publication cutoff and the priority given to primary, peer-reviewed sources.

Planning: inspect the agenda before section research

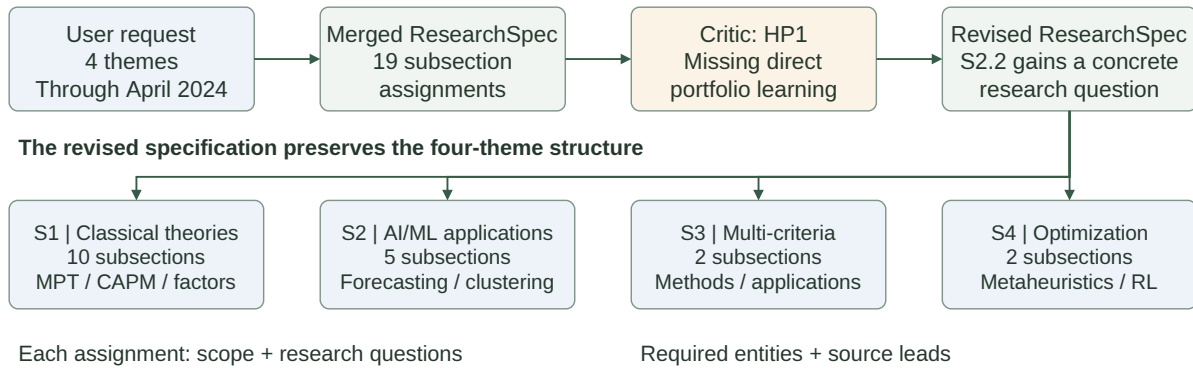


Figure 4: Recorded planning artifacts for the portfolio-management case. The Critic identifies a missing research direction; the Reviser adds it to S2.2 while retaining the 19-subsection structure. The lower row groups the resulting assignments by the user’s four themes. Selected artifacts are shown, not every internal model call.

A concrete revision before research. The merged plan covers forecasting methods but does not name Heaton, Polson, and Witte or “Deep Portfolio Theory.” The Critic flags this omission as HP1 and directs attention to learning portfolio weights from data, alongside the existing forecasting agenda. The Reviser adds the topic to S2.2, including a specific research question and a named entity for subsequent investigation. The change expands an existing assignment; it does not create an additional subsection.

Planning revision	S2.2 Time series forecasting
CRITIC: IDENTIFY THE MISSING DIRECTION HP1: Heaton, Polson & Witte (2017) — “Deep Portfolio Theory”	
REVISER: MAKE THE NEW QUESTION EXECUTABLE How does Heaton, Polson & Witte (2017) “Deep Portfolio Theory” propose learning portfolio weights directly from data, and how does this differ from the predict-then-optimize approach? (HP1)	

From an assignment to a section artifact. The Capital Asset Pricing Model (CAPM) assignment (S1.2) asks which evidence supports or challenges the model and supplies a survey as a source lead. Its recorded Researcher output develops that question into a cited section on empirical tests, Roll’s critique, and Fama–French findings. The final report also contains a dedicated “Deep Portfolio Theory” passage under time series forecasting, preserving the added topic. These are written section artifacts; the next stage coordinates their overlapping material.

5.5.2 Coordinating Evidence Across Section Drafts

The Fama–French findings serve two roles in this report. In the CAPM section they provide an empirical challenge to the model; in the three-factor-model section (S1.5) they motivate an alternative account of returns. Independent section writing can therefore repeat a legitimate piece of evidence. The Global Editor reads the assembled draft and assigns the detailed treatment to S1.5, while directing S1.2 to retain a brief mention and a cross-reference. The Local Editor then applies this decision to the existing CAPM section.

Preserve the finding, coordinate its detailed treatment



Global reads all section drafts. Only the S1.2 editing path is shown.

Figure 5: The selected CAPM editing path. The Global Editor assigns cross-section ownership; the Local Editor receives that directive and the existing section text. The revised passage retains the empirical finding while referring to S1.5 for its detailed treatment. Other sections in the Global Editor’s assembled context are omitted from this view.

Observed text change	S1.2 CAPM
RESEARCH DRAFT Using the Fama-MacBeth methodology on NYSE, AMEX, and NASDAQ stocks (1963–1990), they found that the relation between market beta and average return was flat — beta had no explanatory power for the cross-section of average returns when used alone.	
GLOBAL OWNERSHIP DIRECTIVE Fama-French 1992 empirical findings → S1.5; S1.2 cross-references S1.5	
LOCAL REVISION RETAINED IN THE FINAL REPORT <i>Fama-French (1992).</i> The most damaging empirical challenge to the CAPM came from Fama and French’s (1992) finding that market beta had no explanatory power for the cross-section of average returns, while size and book-to-market equity were strongly and robustly related to average returns (discussed in detail in S1.5).	

What the trace establishes. The plan, draft, directive, and final passage can be linked through the same section identifier. The revision operates on an existing research artifact: it preserves the central empirical finding, condenses its local treatment, and records where the detailed discussion belongs. The Global plan makes similar ownership decisions for the “error maximization” concept and for Non-dominated Sorting Genetic Algorithm II (NSGA-II) details, showing that coordination concerns the allocation of substantive material across sections, as well as sentence-level editing. Those directives are recorded decisions; the CAPM passage above is the specific execution traced here.

Remaining limits. The example also exposes a finalization mismatch: the report retains the internal “S1.5” reference while the displayed three-factor-model heading omits that identifier. The final report still misses some benchmark requirements, including Treynor and CPPI. Thus, an executable research agenda and a traceable edit do not guarantee complete coverage or flawless presentation. Only the final report was scored; the case does not isolate planning or editing gains. Further excerpts and a contrasting labor-market case with unresolved coverage gaps appear in [Section C](#).

6 Conclusion

We presented LongCat-DeepResearch, a model-and-harness system that shifts early research iteration from full reports to ResearchSpec. Parallel planning defines evidence requirements and section responsibilities. Researchers develop citation-bearing sections independently, and coordinated Editors revise the assembled draft. These interfaces also support research-task and trajectory construction.

The system achieves 55.25 on DeepResearchBench, 51.35 on DeepResearchBench II, and 79.83 on ResearchRubrics. On matched development subsets, the full LongCat pipeline has the highest component-study means. Model–harness comparisons favor the current system, although differing evaluation cohorts make those differences descriptive rather than strictly paired estimates. Further planning refinement produces mixed category-level effects, while enhanced Editor rounds improve average readability preference. These observations are specific to the evaluated configurations and do not isolate training-stage contributions.

ResearchSpec coverage supports early inspection of planned requirements; factual verification and independent report evaluation remain necessary. These deep-research improvements are intended to be incorporated into the next general-purpose LongCat model release.

Limitations

Evaluation and automatic assessment. Native benchmark scores, planned coverage, and citation support measure different properties. The in-house evaluation adds a complementary rubric and aggregation protocol, but calibration on a related pairwise task does not establish human agreement for all its dimensions. Trajectory analyses show that unchanged content can receive different rubric judgments. Question-bootstrap intervals capture question variation, not uncertainty from repeated generation or judging. Coverage diagnostics support inspection of planned requirements; factual verification and independent report evaluation remain necessary. Dedicated factual-support and human readability assessments would complement the component studies.

Development evidence and reproducibility. The design analyses use previously inspected development subsets, some selected using prior model scores. Retrieval windows, realized computation, and recovery histories vary across studies; failed-stage recovery retains original attempts and costs. Broader task coverage and repeated runs would help characterize trend stability and resource requirements. Research-task and trajectory construction belong to a broader model development process. The model comparisons evaluate the resulting checkpoints; matched training ablations would help identify contributions from individual data sources and training stages. Filtering, overlap checks, and checkpoint inclusion require separate records; the broader training mixture and optimization recipe belong to the model report. Changing web content and hosted research products also limit exact replay.

Planning and editorial scope. ResearchSpec is refined during planning and fixed during section research. Researchers can investigate further within their assignments, but reopening the global agenda after drafting remains outside the core pipeline. Complete section artifacts support assembly and inspection, yet research and editing can omit relevant details. The framework prioritizes evidence recall and coverage of research requirements. The resulting reports can be lengthy, dense, or repetitive, leaving room to improve readability for human readers. Future work will explore iterative editorial refinement to improve organization and expression while preserving substantive coverage, with human evaluation to assess these trade-offs. Global and Local Editors read the full draft and remain subject to input-context limits. Additional refinement has varying effects across configurations and benchmarks, consuming resources that could also support evidence gathering. The enhanced Editor jointly changes source access, instructions, and iteration count, leaving their individual contributions unresolved.

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A Official Leaderboards and Evaluated Systems

Tables 6 and 7 place the official GPT-5.5 (medium) leaderboard snapshots accessed on September 14, 2026 alongside the four systems evaluated in this work. The lower groups reproduce Table 1; they retain their own system snapshots and evaluation settings and are not official leaderboard entries.

System	Comp.	Insight	Inst.	Read.	Overall
<i>Official leaderboard snapshot (September 14, 2026)</i>					
ATH-Voicepica-DeepResearch	56.37	56.77	55.92	53.27	55.99
cellcog-max	56.34	57.08	55.30	51.94	55.78
SPIA DeepResearch	55.26	55.99	55.12	52.70	55.12
WhaleCloud-DocChain_0612	55.14	55.33	54.85	52.48	54.78
deep-dog-2	54.68	55.32	54.26	51.05	54.32
bodhi	54.15	54.60	54.41	51.87	54.07
lunon_full100_FINAL.submission	53.42	54.83	53.41	50.48	53.51
dalpha-deepresearch	52.58	52.94	53.87	53.20	53.10
sourcery	50.53	52.22	51.06	49.68	51.17
Infosys-MARS-Deepresearch-Agent	50.96	51.79	51.77	48.62	51.12
gemini-2.5-pro-deepresearch	50.01	49.92	50.22	49.58	49.98
openai-deepresearch	48.05	46.69	49.29	47.62	47.84
perplexity-Research	41.78	41.27	45.31	46.03	43.05
grok-deeper-search	39.65	38.12	44.62	45.72	41.22
<i>Systems evaluated in this work</i>					
Gemini-DeepResearch	49.47	50.73	50.88	49.28	50.21
ChatGPT-DeepResearch	55.41	55.82	54.99	51.51	54.95
Claude-DeepResearch	53.71	53.87	53.59	51.35	53.43
LongCat-DeepResearch	56.21	56.48	55.18	49.96	55.25

Table 6: DeepResearchBench RACE results under GPT-5.5 (medium) evaluation. Comp., Inst. and Read. denote comprehensiveness, instruction following and readability. Official rows preserve their published identities; the lower group uses the system snapshots in Section 4.2.

The Gemini row in our evaluation covers 99 scored tasks; the other three rows cover 100. The newer web-product snapshots and their tool budgets differ from those of the published entries.

System	Valid	Info.	Analysis	Pres.	Block	Overall
<i>Official leaderboard snapshot (September 14, 2026)</i>						
GPT-o3 Deep Research	131/132	38.05	46.37	83.49	2.13%	43.00
Gemini 3 Pro Deep Research	129/132	36.58	43.91	84.94	0.71%	41.40
Gemini 2.5 Pro Deep Research	132/132	34.28	49.11	77.98	2.58%	39.98
Doubao Deep Research	127/132	32.58	44.12	71.53	0.92%	37.36
Perplexity Research	127/132	29.16	36.26	73.41	1.04%	33.65
<i>Systems evaluated in this work</i>						
Gemini-DeepResearch	131/132	41.75	51.28	84.81	1.62%	46.72
ChatGPT-DeepResearch	132/132	41.97	52.26	86.98	0.00%	47.16
Claude-DeepResearch	132/132	44.37	51.15	81.02	0.89%	48.18
LongCat-DeepResearch	131/132	46.42	60.69	83.13	0.79%	51.35

Table 7: DeepResearchBench II results under GPT-5.5 (medium) evaluation. Valid denotes scored tasks; Info., Pres. and Block denote information recall, presentation and blocked-source rate. The upper group is the official text-only snapshot; the lower group reproduces our main results.

Our LongCat row uses a 300,000-character Judge-input limit, while the three web-product rows use 150,000 characters. Nineteen LongCat reports exceed the smaller limit. Missing tasks and protocol details remain as stated in [Section B](#); the combined presentation does not establish an official rank or a matched-budget comparison.

Official snapshot provenance. The published entries come from [the official leaderboard](#), Hugging Face Space revision ad8fc70b0e6c, using `data_gpt55/leaderboard.csv` and `data_drb2/leaderboard.csv`. The latter is the GPT-5.5 text-only reevaluation of public reports, excluding PDF-only submissions and ignoring embedded DOCX images. The dated snapshot is retained unchanged here.

B Experimental Details and Supplementary Results

B.1 Main Evaluation and Shared Conventions

The LongCat DeepResearchBench result covers 100 scored tasks; ResearchRubrics covers 101 tasks and 2,593 criteria, with thinking enabled, three planning writers, twenty maximum tool rounds and research concurrency three. The selected DeepResearchBench II result uses 131 scored reports out of 132 assigned tasks, excluding missing task 116. Its overall and dimension scores share this denominator. The 300,000-character Judge-input limit retains nineteen reports longer than 150,000 characters without truncation; the listed web products use a 150,000-character limit.

Web-product coverage also varies: Gemini-DeepResearch has 99 scored DeepResearchBench tasks and 131 DeepResearchBench II tasks, whereas ChatGPT-DeepResearch and Claude-DeepResearch have 100 and 132, respectively. ChatGPT-DeepResearch and Gemini-DeepResearch ResearchRubrics scores use their common 100-task cohort. Claude-DeepResearch’s ResearchRubrics overall is an externally reported aggregate whose complete original grades are unavailable in the inspected archive. These cohorts and input limits define the scope of the point estimates in [Table 1](#).

Native scoring and recovery. DeepResearchBench uses the benchmark’s GPT-5.5 (medium) RACE evaluator. The development DeepResearchBench II studies use GPT-5.5 (medium), 50-rubric chunks, a 150,000-character input cap and the complete rubric denominator. ResearchRubrics uses Gemini 2.5 Pro, binary verdicts, signed criterion weights and the sum of positive weights as denominator. Recovery retains completed report bytes and grades, reuses exact matching responses, and charges failed attempts to the original task. Successful low-score reports are not selectively regenerated. These measurements include recovery rather than estimating first-attempt reliability.

In-house evaluation. GPT-5.6 Sol (xhigh) evaluates the in-house comparison using question-specific pointwise rubrics and pairwise comparisons in both report orders. The five dimension scores and weighted overall results are reported in [Section 4.5](#). Order reversal addresses presentation sensitivity; calibration on a related RACE preference task does not independently validate this five-dimension protocol.

B.2 Component Comparison

LongCat uses pro-v-int8-v082902, with thinking enabled. The comparison uses fixed ResearchRubrics and DeepResearchBench II development subsets, restricted to questions completed by all four arms. An incomplete single-Researcher case is excluded from the paired comparison; its completed outcomes remain in the records. The interventions are defined in [Section 5.1](#). The single Researcher’s tool allowance sums the section allowances within the task ceiling, and memory compression is disabled in the component variants.

None of the included LongCat component reports exceeds the scorer’s 150,000-character input cap. The studies have one completed grade per report and do not estimate repeated-Judge variance. Official reference-result alignment remains unestablished for these previously inspected development subsets.

B.3 Model-Harness Configuration Settings

The baseline configurations in [Table 5](#) use fixed development subsets calibrated to prior scores; the displayed current-system row reports the full-cohort main results. ReAct/direct reporting has a 65,536-token final-response ceiling, while the current harness in this comparison has a 32,000-token per-call ceiling. Cumulative task limits are 600 model calls, 1,048,576 completion tokens, 500 searches and 500 page reads. These allowances do not imply equal realized compute. Same-model recovery retains validated plans and sections, original spending and unchanged scoring rules. Comparisons across retrieval windows and recovered runs characterize the recorded configurations.

B.4 ResearchRubrics Refinement Protocol

[Table 3](#) uses the same LongCat development subset across all four stages. For each question, W0 reuses the first Writer from a single three-Writer planning execution; it is not selected by report score. J0 is the Judge’s integrated plan before critique or revision. R1 revises J0 once, and R2 revises R1 once more. The experiment disables planning compaction and explicitly executes both Critic/Reviser cycles. Each of the four immutable planning snapshots is passed to the original parallel Researchers and Editor. Generation uses thinking and seed 42 with the original cumulative task budgets. Comparisons use matched question IDs, and results are averaged over questions.

Fast ResearchSpec coverage. Before report generation, the existing GPT-5.5 (medium) diagnostic can assess the ResearchSpec alone against the same question’s rubric. It maps each criterion to plan excerpts and section identifiers,

without executing Researcher or Editor. Coverage is the percentage of positive-weight criteria judged fully covered, with criteria counted equally rather than weighted. Partial, missing and not-plan-evaluable criteria remain in the denominator. We average the per-question percentages over the same development subset. This is a planning diagnostic, distinct from native report quality or factual verification; all entries reuse existing judgments.

Explicit and implicit requirements. We group the original ResearchRubrics criteria by their supplied *Explicit Criteria* and *Implicit Criteria* labels. For each question and category, the score is 100 times the sum of satisfied criteria’s signed weights divided by the sum of positive weights in that category. The table reports the mean of these per-question scores; both categories are present throughout the development subset. Negative-weight criteria retain their penalties. Total uses all rubric categories and their full positive-weight denominator, so it is not the average of the two displayed subscores. The native Gemini 2.5 Pro verdicts are reused for this breakdown; category aggregation introduces no additional Judge calls. Rubrics and diagnostic judgments are never supplied to the generating agents.

B.5 Historical Planning-Budget Study: Cost and Uncertainty

This earlier three-level study is separate from the four-stage ResearchRubrics comparison in Table 3; its costs and intervals do not describe that newer experiment. The three levels start from the same completed LongCat ResearchSpec. Zero extra cycles already includes the original multi-planner, consolidation and Critic/Reviser pipeline; the other levels add one or two Critic/Reviser cycles. Each level independently runs Researcher and Editor with thinking enabled and a 64,000-token output ceiling. Cumulative limits are 600 calls, 1,048,576 completion tokens, 500 searches and 500 page reads. Each branch inherits planning costs, but not another branch’s report-generation costs. The Spec remains fixed during research, and benchmark rubrics or diagnostic scores do not enter the Actor inputs.

Plan coverage. The offline GPT-5.5 (medium) diagnostic maps criteria to exact plan excerpts and valid section identifiers, using chunks of at most fifty criteria and a 32,000-token output ceiling. Coverage is the unweighted fraction of positive-weight criteria judged fully covered; partial, missing and not-plan-evaluable items remain in the denominator. Across zero, one and two extra cycles, coverage is 34.34/33.87/32.84% on DeepResearchBench II and 66.42/67.07/67.08% on ResearchRubrics. This diagnostic makes planned requirements inspectable but does not verify facts or provide a calibrated selector of the best report iteration.

Resource use and paired outcomes. Table 8 includes inherited planning and failed-stage recovery costs; runtime excludes idle gaps between recovery campaigns. Shared prefixes are attributed to each logical report without counting them again as new physical requests. Recovery retains completed plans, research sections, Global directives and accepted edits. One DRB-II Local repair uses an exact-span citation restoration followed by the original whole-section checks; successful reports and grades remain unchanged.

Benchmark	Extra	Planning		Total generation			Spec scoring
		Calls	Output (k)	Calls	Output (k)	Min.	Min.
DeepResearchBench II	0	28.0	106.0	98.9	215.4	59.16	1.37
DeepResearchBench II	1	33.5	138.7	104.6	263.6	61.36	1.10
DeepResearchBench II	2	38.6	173.0	110.6	294.1	67.82	1.04
ResearchRubrics	0	29.2	108.6	107.1	249.5	65.13	0.76
ResearchRubrics	1	34.7	145.7	113.8	301.4	70.65	0.93
ResearchRubrics	2	39.9	181.7	116.5	344.0	77.58	0.67

Table 8: Mean recorded resource use per logical report. Extra counts additional Critic/Reviser cycles; Calls counts model requests; Output (k) is output tokens in thousands; Min. is elapsed minutes. Spec scoring evaluates ResearchSpec coverage.

Intervals in Table 9 use 10,000 paired question-bootstrap draws with seed 1729. They describe question variation, without estimating repeated-generation or repeated-Judge uncertainty. The native-score response is nonmonotonic: ResearchRubrics peaks after one extra cycle, while DeepResearchBench II declines across the three levels. The development-set selection, retrieval windows and recovery histories remain part of these observations.

Benchmark	Extra cycles	Difference (points)	95% interval
DeepResearchBench II	1 vs. 0	-1.38	[-5.98, 2.94]
DeepResearchBench II	2 vs. 0	-4.19	[-8.42, -0.36]
ResearchRubrics	1 vs. 0	2.50	[-0.89, 5.74]
ResearchRubrics	2 vs. 0	0.34	[-2.45, 3.00]

Table 9: Question-paired native-score changes relative to zero extra planning cycles.

B.6 Editor Refinement: Settings and Paired Results

LongCat uses fixed development subsets for each benchmark and five conditions. The model identifier is `pro-v-int8-v082902`. Thinking is enabled and seed 42 is requested. The development subsets were expanded after inspection of the initial results; this is exploratory rather than independent confirmation.

A0 is the unedited draft. A1 applies the original Editor once; B1 applies the enhanced Editor once, independently from the same A0. B2 and B3 apply the second and third enhanced rounds, each continuing its predecessor. The enhanced Global Editor receives citation-bound excerpts from already retrieved sources, limited to 24,000 characters in total and 2,400 per source, without new retrieval. Source access, fact-checking instructions and iteration count form a joint intervention. Per-call output permits 64,000 tokens; cumulative task limits match the planning study. Failed stages retain original costs and accepted artifacts. LongCat recovery includes directive-bound URL correction and syntax repair in auxiliary quotation fields, preserving native verdicts and scores.

The native scoring rules follow the shared conventions above. Some LongCat DRB-II reports exceed the 150,000-character cap; the affected reports belong to the same question. ResearchRubrics receives full reports. Each condition has one completed grade; official reference-result alignment remains unestablished.

Average readability preference. We reuse the original portable organization-and-expression evaluator with GPT-5.5 (medium), thinking enabled and seed 42. Each edited report is compared with its own A0 draft using full reports, three guides and both report orders. For each question, we average the guide margins after orienting both orders toward the edited report, preserving the original guide weighting. A margin $m \in [-2, 2]$ maps to $50 + 25m$; 50 denotes parity with A0 rather than an absolute readability level. The displayed mean weights the two benchmark subsets equally, giving the unweighted mean of the two benchmark means. A0 has no self-judgment. Scores are automatic preference judgments rather than human annotations; original judgments and quotation warnings remain in the experiment records.

The aggregate trend need not hold within each benchmark. For example, LongCat B3 has mean margins of -0.183 on DRB-II and $+0.500$ on ResearchRubrics relative to A0, yielding $+0.158$ overall. Adjacent-round preferences cannot be added to obtain preference over A0; every displayed condition is evaluated directly against A0.

Paired results. Table 10 reports the paired contrasts supporting Table 4. Intervals use 10,000 question-bootstrap draws with seed 42 and are unadjusted for multiple comparisons. LongCat B3–A0 changes DRB-II by $+2.51$ and ResearchRubrics by $+0.09$. These intervals cover question variation, not repeated-generation or repeated-Judge variance.

Benchmark	Difference	Mean change	Descriptive 95% interval
DRB-II	A1 – A0	+0.17	[-0.71, +1.11]
DRB-II	B3 – A0	+2.51	[+0.63, +4.78]
DRB-II	B1 – A1	-0.89	[-2.51, +0.56]
DRB-II	B2 – B1	+0.52	[-0.57, +1.70]
DRB-II	B3 – B2	+2.70	[+0.51, +5.24]
ResearchRubrics	A1 – A0	-0.25	[-2.08, +1.89]
ResearchRubrics	B3 – A0	+0.09	[-0.79, +1.18]
ResearchRubrics	B1 – A1	-0.48	[-3.00, +1.89]
ResearchRubrics	B2 – B1	+1.69	[-0.03, +3.59]
ResearchRubrics	B3 – B2	-0.87	[-2.35, +0.58]

Table 10: LongCat paired native-score changes on fixed development subsets. A0 is the unedited draft, A1 one original Editor pass, and B1–B3 one to three enhanced Editor rounds. Positive changes favor the first condition. DRB-II denotes DeepResearchBench II.

What the edits change. On DRB-II 58, editing corrects the online-learning study table from “Arab Region” and 3,348 participants to “Jordan” and “280 students + 50 faculty,” consistent with the detailed discussion. The native score rises by 1.59 points. On the universal basic income task, clearer pilot-specific sections gain a structural criterion, but weaker attribution of quantitative claims loses a higher-weight citation criterion, yielding -1.25 points. These examples retain both a bounded repair and a content-preservation cost.

Some changes instead reflect Judge sensitivity: most newly credited hydroforming facts were already in the unedited draft (A0). Original grades are retained. Report diffs and the native scores therefore provide distinct views of editorial changes. No model parameters are updated in this study.

C Case Studies: From ResearchSpec to Edited Report

English-language examples from the selected DeepResearchBench II cohort. GPT-5.5 (medium); 300k-character input cap; neither report truncated.

In the excerpts, $Sx.y$ identifies subsection y of section x ; HP1 labels the Critic's first listed issue. MPT denotes modern portfolio theory, CPPI constant-proportion portfolio insurance, and CI confidence interval.

CASE 01 AI-enhanced portfolio management

DeepResearchBench II 012

FINAL SCORE **89.29 / 100**

TASK & SCOPE

Review classical theories, AI/ML applications, multi-criteria decision making, and portfolio optimization and rebalancing through April 2024.

Task synopsis. The complete ResearchSpec assigns 19 subsections.

RESEARCHSPEC

research_spec.md / ReportSpec / selected original passages

Global research boundaries

Output language: English (matching the research question).

[...]

S1.2 | Capital Asset Pricing Model (CAPM)

Within S1: Classic Theoretical Frameworks

What to Cover

Present CAPM as the equilibrium asset pricing model building on MPT. Explain the core idea: expected excess return of any asset is proportional to its beta (systematic risk) relative to the market portfolio.

[...]

Research Questions

Who originated CAPM and in what years (Sharpe 1964, Lintner 1965, Mossin 1966)?

[...]

Required Entities / Cases

[...]

Jan Mossin (1966)

Security Market Line (SML)

[...]

Source Leads

<https://www.investopedia.com/terms/c/capm.asp> - CAPM overview (orientation only; find primary peer-reviewed source)

[...]

FINAL EVALUATION

The plan still omits Treynor (1962); the final report also misses CPPI and fails the chronological-order criterion.

Source text is abridged, not rewritten. Field labels and Markdown are typeset for readability; [...] marks omitted passages. The panels show selected ResearchSpec fields and source passages.

CASE 01 From the plan to the edited report

Recorded stage changes

DeepResearchBench II 012

01 PLANNING REFINEMENT**Critic identifies a missing method**

HP1: Heaton, Polson & Witte (2017) — "Deep Portfolio Theory"

Reviser adds a research question in S2.2

How does Heaton, Polson & Witte (2017) "Deep Portfolio Theory" propose learning portfolio weights directly from data, and how does this differ from the predict-then-optimize approach? (*HP1*)

Heaton et al. is absent from the merged spec and present in the revised spec. This changes the research agenda before section writing.

02 RESEARCH DRAFT

S1.2 / CAPM researcher / excerpt before editing

Using the Fama-MacBeth methodology on NYSE, AMEX, and NASDAQ stocks (1963–1990), they found that the relation between market beta and average return was **flat** — beta had no explanatory power for the cross-section of average returns when used alone.

03 GLOBAL EDIT

Ownership decision after the section drafts are assembled

Fama-French 1992 empirical findings → S1.5; S1.2 cross-references S1.5

04 LOCAL EDIT → FINAL REPORT

S1.2 / revised passage

Fama-French (1992). The most damaging empirical challenge to the CAPM came from Fama and French's (1992) finding that market beta had no explanatory power for the cross-section of average returns, while size and book-to-market equity were strongly and robustly related to average returns (discussed in detail in S1.5).

This passage survives in the scored final report. The CAPM discussion keeps the principal finding and points to S1.5 for the detailed treatment.

WHAT CHANGED

The plan gains a specific research question. Section research expands assignments into cited prose. Editing coordinates repeated material across sections through an ownership decision and a shorter cross-reference.

What this does not establish

Only the final report has a benchmark score. This trace does not measure the score contribution of planning or editing.

Quotes retain the recorded wording. Commentary describes observed text changes, not measured intermediate score gains.

CASE 02 Generative AI and the labor market

DeepResearchBench II 056

FINAL SCORE **11.43 / 100****TASK AND RESEARCH AGENDA**

Review pre-June 2023 literature on positive views, negative views, challenges and opportunities, followed by a study-level application table. The ResearchSpec contains fourteen subsections across these themes and a separate summary-table section.

CRITIC → REVISED SPEC

HP1: Peng et al. (2023) — GitHub Copilot Developer Productivity Study

The merged Spec does not name Peng et al. or GitHub Copilot. The revised Spec adds the study, its arXiv source lead, and a research question:

What productivity effect did Peng et al. (2023) document for software developers using GitHub Copilot, and what does the heterogeneous-effects finding suggest about career transitions into software development? [HP1]

RESEARCH → FINAL REPORT

The treatment group completed the task **55.8% faster** than the control group (71.17 minutes vs. 160.89 minutes on average, $p = 0.0017$, 95% CI: 21–89%).

This sentence appears in the Researcher output and survives editing in the scored final report. Global editing also assigns the aggregate labor-exposure figures to S2.1, and the Local Editor applies that cross-reference.

REMAINING COVERAGE GAP

The 121,366-character final report emphasizes aggregate labor-market exposure and prominent productivity studies. It omits the rubric-required studies *Can AI help for scientific writing?* and *Comparing physician and AI chatbot responses to patient questions posted to a public social media forum*. Both titles are absent from the revised Spec and final report, and information recall remains zero.

TAKEAWAY

The planning addition and its downstream passage are traceable, while the remaining coverage gap persists. This case illustrates how refining an existing agenda can leave other required evidence uninvestigated; its final score does not isolate the contribution of an individual stage.

Quotes are abridged from recorded artifacts. Final-report hashes match the original grades; full Specs, stage outputs and exact spans remain in the source records.

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